

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	7 July 2026
Roll Number	23A51A4265
Project Name	Leave Approval System
Maximum Marks	4 Marks

Technical Architecture (Leave Approval System)

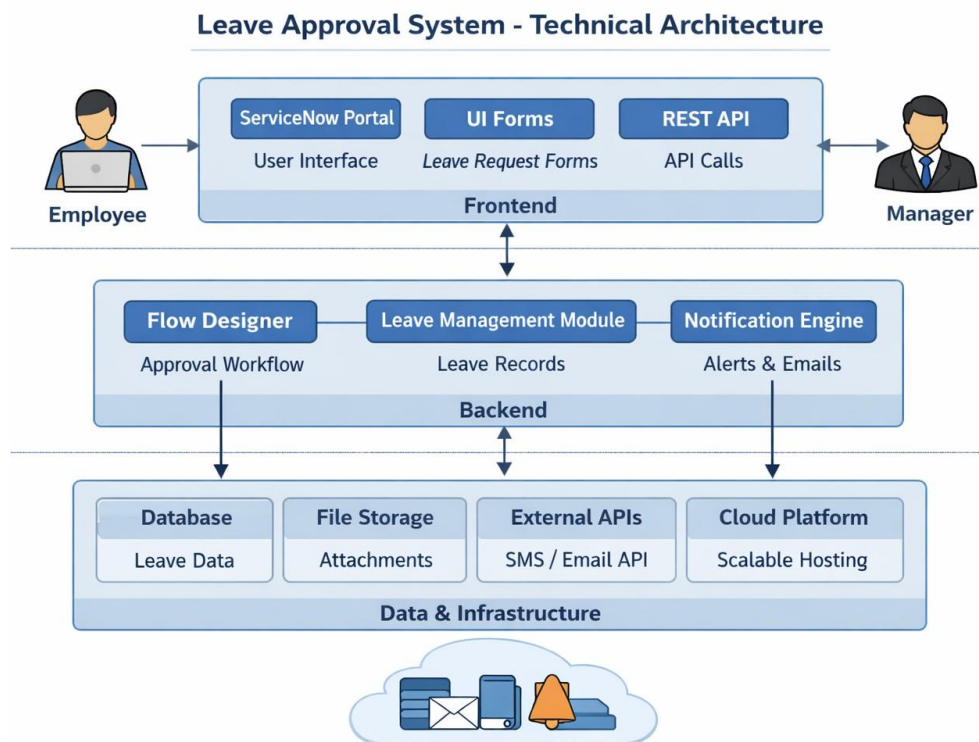


Table 1: Components

S.No	Component	Description	Technology
1	User Interface	Employee interacts via Web UI or Mobile App	HTML, CSS, JavaScript, Angular/React
2	Application Logic-1	Leave request submission & validation (end $\geq$ start date)	JavaScript / ServiceNow Scripting
3	Application Logic-2	Workflow routing for manager approval	ServiceNow Flow Designer
4	Application Logic-3	Notification logic for approval/rejection	ServiceNow Notification Engine

S.No	Component	Description	Technology
5	Database	Stores leave requests, approvals, and reports	MySQL / ServiceNow Tables
6	Cloud Database	Cloud storage for scalability	ServiceNow Cloud DB / AWS RDS
7	File Storage	Stores attachments (medical certificates, etc.)	ServiceNow Attachments / Cloud Storage
8	External API-1	Email/SMS notification integration	Twilio API / SMTP
9	External API-2	HR system integration for employee records	REST API
10	Infrastructure	Deployment on ServiceNow Cloud	ServiceNow PaaS / Kubernetes

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	UI frameworks for responsive design	Angular, React
2	Security Implementations	Role-based access, encryption, IAM controls	SHA-256, HTTPS, ServiceNow ACLs
3	Scalable Architecture	Microservices & cloud-based deployment	ServiceNow Cloud, Kubernetes
4	Availability	High availability with load balancing	ServiceNow Cloud, AWS Load Balancer
5	Performance	Optimized workflows, caching, fast notifications	ServiceNow Performance Analytics, CDN